

Business Requirements Document (BRD) - ShuttleIQ

Version: 1.0

Purpose: Define business requirements for ShuttleIQ, an AI-powered badminton training and tournament management platform for India.

Executive Summary

ShuttleIQ is a web, mobile, and desktop platform focused on tournament organizers while serving players, coaches, academies, and parents. The system supports BWF-compliant tournaments and AI-driven player development.

Business Objectives

- Simplify badminton tournament operations.
- Provide player training and performance insights.
- Enable BWF-compliant fixture generation.
- Offer nutrition, warm-up, recovery, and injury prevention guidance.
- Build a nationwide badminton ecosystem.

Stakeholders

Primary: Tournament Organizers.

Secondary: Players, Coaches, Academies, Parents.

Tertiary: Umpires, Referees, Sponsors, Administrators.

Scope

In Scope:

- Tournament Management
- Fixture Engine
- Player Profiles
- Rankings
- Match Statistics
- AI Style Analyzer
- Training Planner

- Diet & Recovery
- Notifications

Out of Scope (Phase 1):

- Wearable integrations
- Marketplace
- International federations

Functional Requirements

FR1: Create and manage tournaments.

FR2: Support Singles, Doubles, Mixed Doubles (Men/Women).

FR3: Generate fixtures using Knockout, Round Robin, Swiss and Hybrid formats.

FR4: Maintain player rankings.

FR5: Record match scores and statistics.

FR6: Provide AI-based training recommendations.

FR7: Generate player performance reports.

FR8: Send notifications and reminders.

Tournament Requirements

- BWF rules compliance.
- Seeding support.
- Court allocation.
- Referee assignment.
- Live score updates.
- Certificates and reports generation.

Player Module

Player profile management.

Match history.

Statistics dashboard.

Training recommendations.

Nutrition guidance.

Warm-up and recovery routines.

Coach Module

Monitor player progress.
Assign training plans.
Review videos and performance.
Provide feedback.

AI Features

- Playing Style Analysis.
- Personalized Training Plans.
- Match Outcome Prediction.
- Video Analysis.
- Opponent Analysis.
- Injury Risk Prediction.
- AI Coach Assistant.

Non-Functional Requirements

Availability: 99.5%
Security: Role-based access control.
Performance: Fixture generation under 30 seconds.
Scalability: Support 100,000+ users.
Auditability: Log all critical actions.

Technology Stack

Frontend: React, Flutter, Electron.
Backend: .NET 9 APIs.
Database: PostgreSQL.
Caching: Redis.
Infrastructure: Docker, Kubernetes, Azure.

Success Metrics

50,000 players.
500 organizers.
2,000 tournaments.

100,000 matches managed.

25,000 monthly active users.

Release Roadmap

Release 1: Tournament Engine, Rankings, Notifications.

Release 2: AI Training, Diet, Coach Dashboard.

Release 3: Video Analytics and Injury Prediction.

Approval

Role	Status
Product Owner	Pending
Business Sponsor	Pending
Technical Lead	Pending